

Business Meeting



2024 TAIPEI

Quantum Information Processing 2024

第二十七屆國際量子資訊處理年會



TAIPEI, TAIWAN

Date: January 13 - 19, 2024

Venue: Taipei International Convention Center (TICC), Taipei

QIP 2024 Steering Committee

Laura Mančinska (Chair) | University of Copenhagen

David Gosset (Co-chair) | University of Waterloo, Perimeter Institute

Omar Fawzi | ENS Lyon

Min-Hsiu Hsieh | Hon Hai (Foxconn) Quantum Computing Research Center

Elham Kashefi | CNRS, University of Edinburgh

Barbara Kraus | University of Innsbruck

Troy Lee | University of Technology Sydney

John Smolin | IBM Research

Frank Verstraete | Ghent University

Thomas Vidick | California Institute of Technology

Agenda

- Report from the Local Organizing Committee
- Report from the Program Committee
- QIP 2025 (Duke)
- Bids for future QIPs (Latvia and Singapore)
- Discussion

2024 TAIPEI

A large, light blue background graphic featuring the letters 'QAP' in a stylized font. The 'Q' contains a map of Taiwan, and the 'P' is partially obscured by the Taipei 101 tower. The text '2024 TAIPEI' is faintly visible at the bottom.

A report from the Local Organizing Committee

Hao-Chung Cheng
(National Taiwan University)

2024 TAIPEI

Sponsors



Diamond



Quantum AI



Gold

IBM Quantum



Silver

amazon | science



JPMORGAN CHASE & CO.

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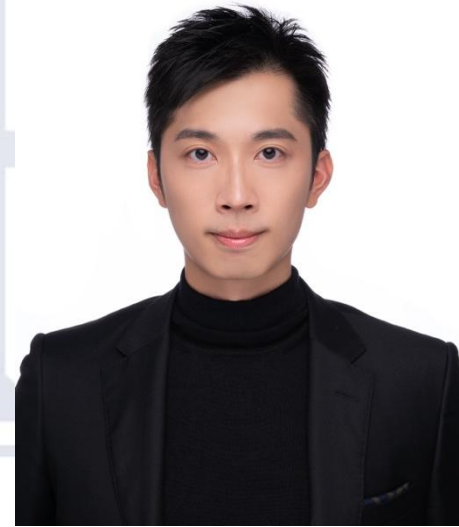


Local Organizing Committee (1/2)

- **Chair**



Min-Hsiu Hsieh 謝明修
(Hon-Hai Quantum Computing Center)



Hao-Chung Cheng 鄭皓中 (NTUEE)



Local Organizing Committee (2/2)

- Member



Ching-Ray Chang 張慶瑞

(CYCU-Phys)



His-Sheng Goan 管希聖 (NTU-Phys)



國立臺灣大學
National Taiwan University



Kai-Min Chung 鍾楷閔

(Academia Sinica-IIS)



Yeong-Cherng Liang 梁永成
(NCKU-Phys)



國立成功大學
National Cheng Kung University



Kuei-Lin Chiu 邱奎霖 (NSYSU-Phys)



國立中山大學
National Sun Yat-sen University



Ching-Yi Lai 賴青沂 (NYCU-EE)



國立陽明交通大學
NATIONAL YANG MING CHIAO TUNG UNIVERSITY



Han-Hsuan Lin 林瀚仟 (NTHU-CSIE)



國立清華大學
NATIONAL TSING HUA UNIVERSITY

Assistant Team

- Financial



Angel Huang 黃綉茹



- PCO



Ting Yang 楊怡婷、Elle Lin 林思樺



- Assistant



Yi-Shan Liu 劉怡嫻



國立臺灣大學
National Taiwan University

47 Student Workers

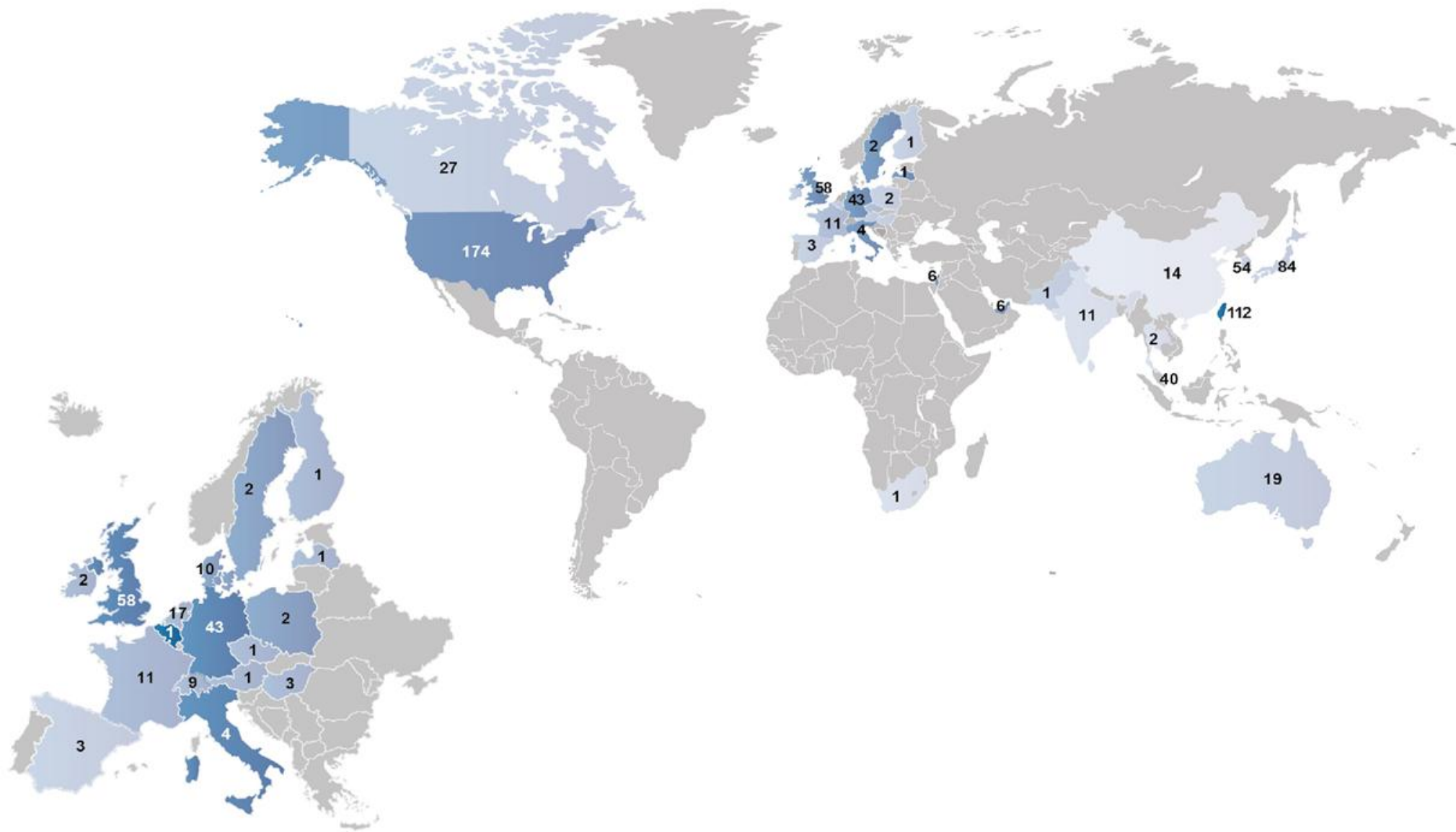
Chen-Chia Lee	Hsuan Yu	Ssu-Yi Chen	Shih Xian Yang	Ching-Wen Huang	Cheng-Han Liu
Chia-Yi Chou	Mu-En Liu	Leo Chiang	Wei-Chen Huang	Cian-Jhu Chen	Tian-Fu Chen
Bo-Yu Yang	Chun-Yuan Hsu	Tzu-Chi Yin	Hsin-Yu Hsu	Chung-Ting Tsai	Ta-Min Chien
Kuan-Hao Chiao	Kuan-Jen Wang	Yu-Hung Pan	Wan-Guan Chang	Che-Ming Chang	Chien-Kai Ma
Ting-Hsun Chi	Po-Shun Chiu	Wei-en Chiu	Yun-Che Lin	Hsin-Juei Sun	Chin-Yi Lin
Cheng-Ze Jiang	Po-Hung Chen	Yu-Chao Huang	ChiAn Lin	Chien-Hung Cho	Chun-Tse Li
Shu-Wen Chen	Chin-Yi Cheng	Jian-Cheng Tzeng	Zong-Tzung Yang	Yen-Chi Lee	Yi-Wen Chen
Jyun-Ting Liu	Yu-Wei Fan	Meng-Zhe Zhong	Ming-Chien Hsu	Jin-Yuan Zheng	

Statistics

- **Registered participants: 734**
- **Students: (396) ~ 54%**

2024 TAIPEI

Australia	19
Austria	1
Belgium	1
Canada	27
China	14
Czech Republic	1
Denmark	10
Finland	1
France	11
Germany	43
Hong Kong	14
Hungary	3
India	11
Ireland	2
Israel	6
Italy	4
Japan	84
Korea	54
Latvia	1
Netherlands	17
Pakistan	1
Poland	2
Singapore	40
South Africa	1
Spain	3
Sweden	2
Switzerland	9
Taiwan	112
Thailand	2
UK	58
United Arab Emirates	6
USA	174



Budget (estimated)

In:

Registration	318k USD
Sponsoring	206k USD
QIP2023	21k USD
Total	545k USD

* Student Stipend

Waive all registration fee and partial travel support based on individual cases (with complete documents)

Out:

Venues	156k USD
Catering (coffee,...)	77k USD
Banquet	37k USD
Invited speakers	5k USD
Admin	101k USD
Misc	44k USD
* Student stipend	61k USD

Total 481k USD

Previous QIPs:

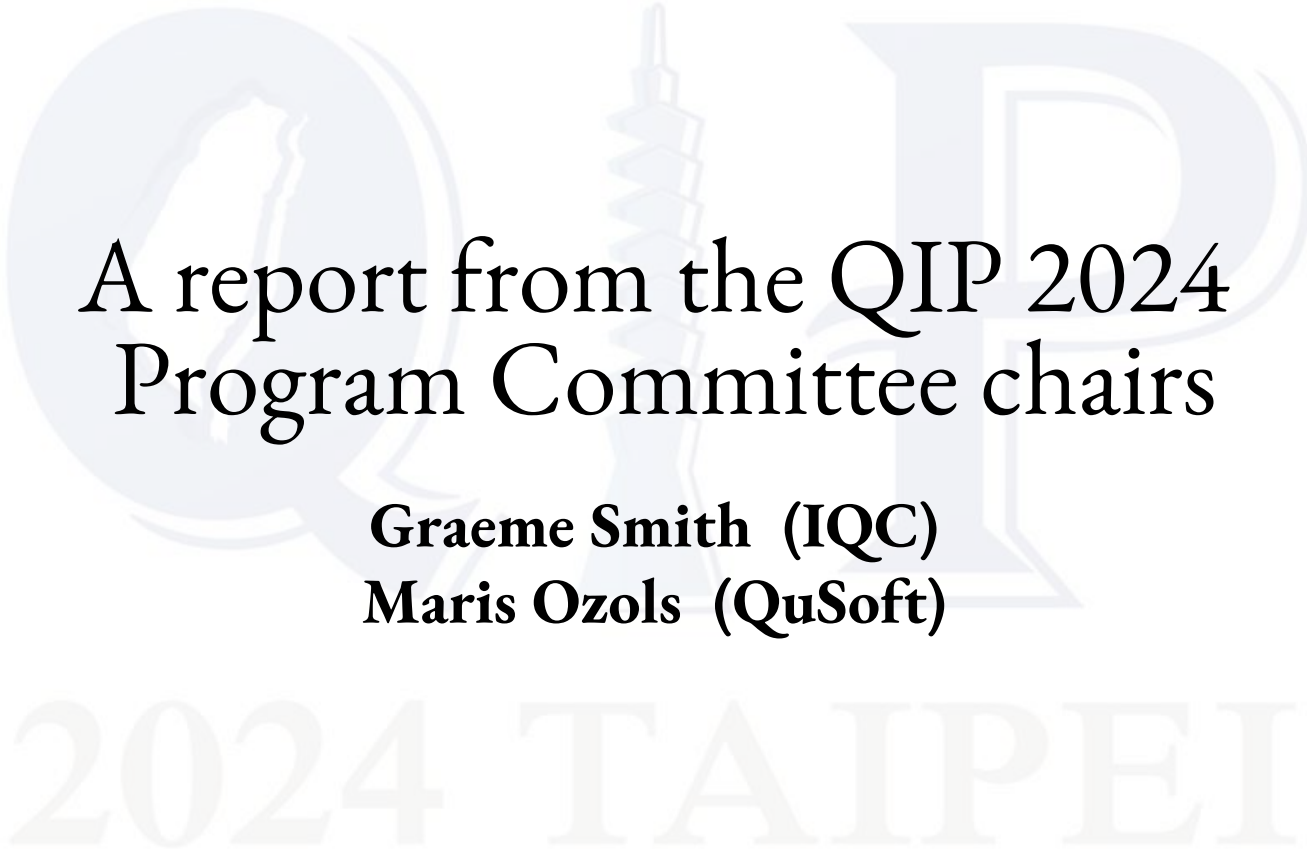
QIP 2023 (Ghent):	~620kUSD
QIP 2022 (Caltech):	~750k USD
QIP 2021 (Munich, online):	~164k EUR
QIP 2020 (Schenzen):	~450k USD
QIP 2019 (Boulder):	~300k USD

One new thing in QIP 2024



Jan. 14 (Sun.)	Jan.15 (Mon.)	Jan. 16 (Tue.)	Jan. 17 (Wed.)	Jan. 18 (Thu.)
Poster Display Session	Poster Presentation Session	Poster Presentation Session	Banquet	Poster Display Session

2024 TAIPEI



A report from the QIP 2024 Program Committee chairs

Graeme Smith (IQC)
Maris Ozols (QuSoft)

Program committee of 66

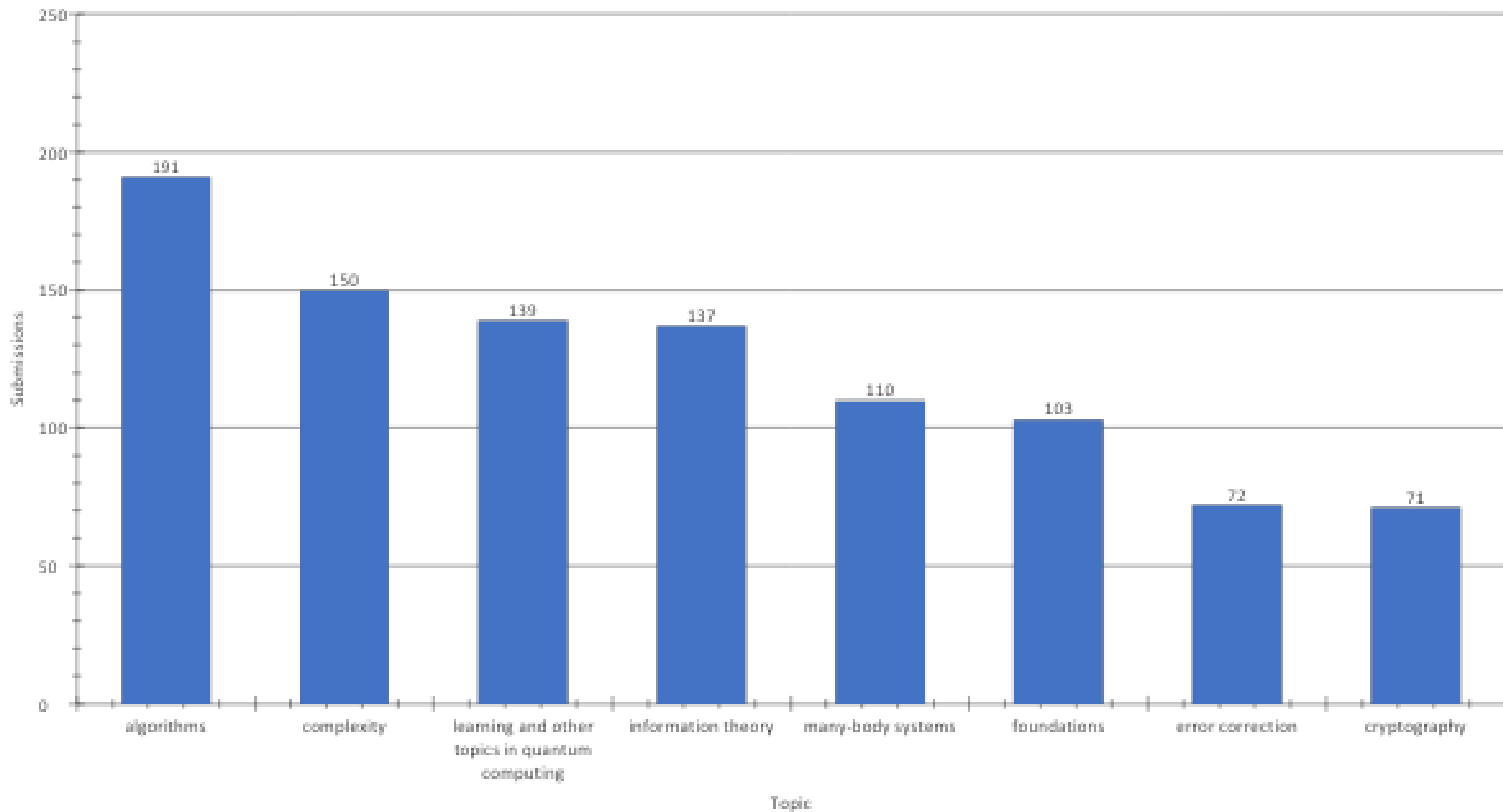
- Gerardo Adesso (Nottingham)
- Victor Albert (NIST and UMD)
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- Shalev Ben-David (IQC)
- Robin Blume-Kohout (Sandia)
- Adam Bouland (Stanford)
- Michael Bremner (UTS)
- Nikolas Breuckmann (Bristol)
- Jacob Bridgeman (Ghent)
- Anne Broadbent (Ottawa)
- Benjamin Brown (IBM)
- Andrew Childs (UMD)
- Eric Chitambar (UIUC)
- Matthias Christandl (QMATH)
- Lidia Del Rio (Squids)
- Di Fang (Duke)
- Steve Flammia (AWS)
- Christopher Fuchs (UMass Boston)
- Ivette Fuentes (Southampton)
- Xun Gao (JILA)
- Craig Gidney (Google)
- Daniel Grier (UCSD)
- Alex Grilo (CNRS)
- Jonas Helsen (QuSoft)
- Pawel Horodecki (Gdansk)
- Hsin-Yuan Robert Huang (Google/MIT)
- Marcus Huber (TUW and IQOQI-V)
- Zhengfeng Ji (Tsinghua)
- Robert Koenig (Munich)
- Tristan Kraft (Innsbruck)
- William Kretschmer (Berkeley)
- Alex Kubica (Yale)
- Ludovico Lami (QuSoft)
- Felix Leditzky (UIUC)
- Ke Li (Harbin)
- Prabha Mandayam (IIT Madras)
- Xiongfeng Ma (Tsinghua)
- Alex May (Perimeter)
- Ashley Montanaro (Phasecraft and University of Bristol)
- Alexander Müller-Hermes (University of Oslo)
- Chinmay Nirkhe (IBM)
- Maris Ozols (QuSoft) Co-chair
- Anna Pappa (Berlin)
- Hakop Pashayan (FU Berlin)
- David Perez Garcia (ICMAT)
- Shruti Puri (Yale)
- Yihui Quek (MIT)
- Cambyse Rouze (Inria and IPP)
- Luke Schaeffer (UMD)
- Pranab Sen (TIFR)
- Peter Shor (MIT)
- Vikesh Siddhu (IBM)
- Graeme Smith (IQC) Chair
- Daniel Stilck França (ENS de Lyons)
- Barbara Terhal (Delft)
- Julio de Vicente (ICMAT)
- Adam Bene Watts (IQC)
- Dominic Williamson (Sydney)
- Andreas Winter (Barcelona)
- Freek Witteveen (QMATH)
- Penghui Yao (Nanjing)
- Kevin Young (Sandia)
- Henry Yuen (Columbia)
- Bei Zeng (HKUST)
- Sisi Zhou (Perimeter)

Statistics

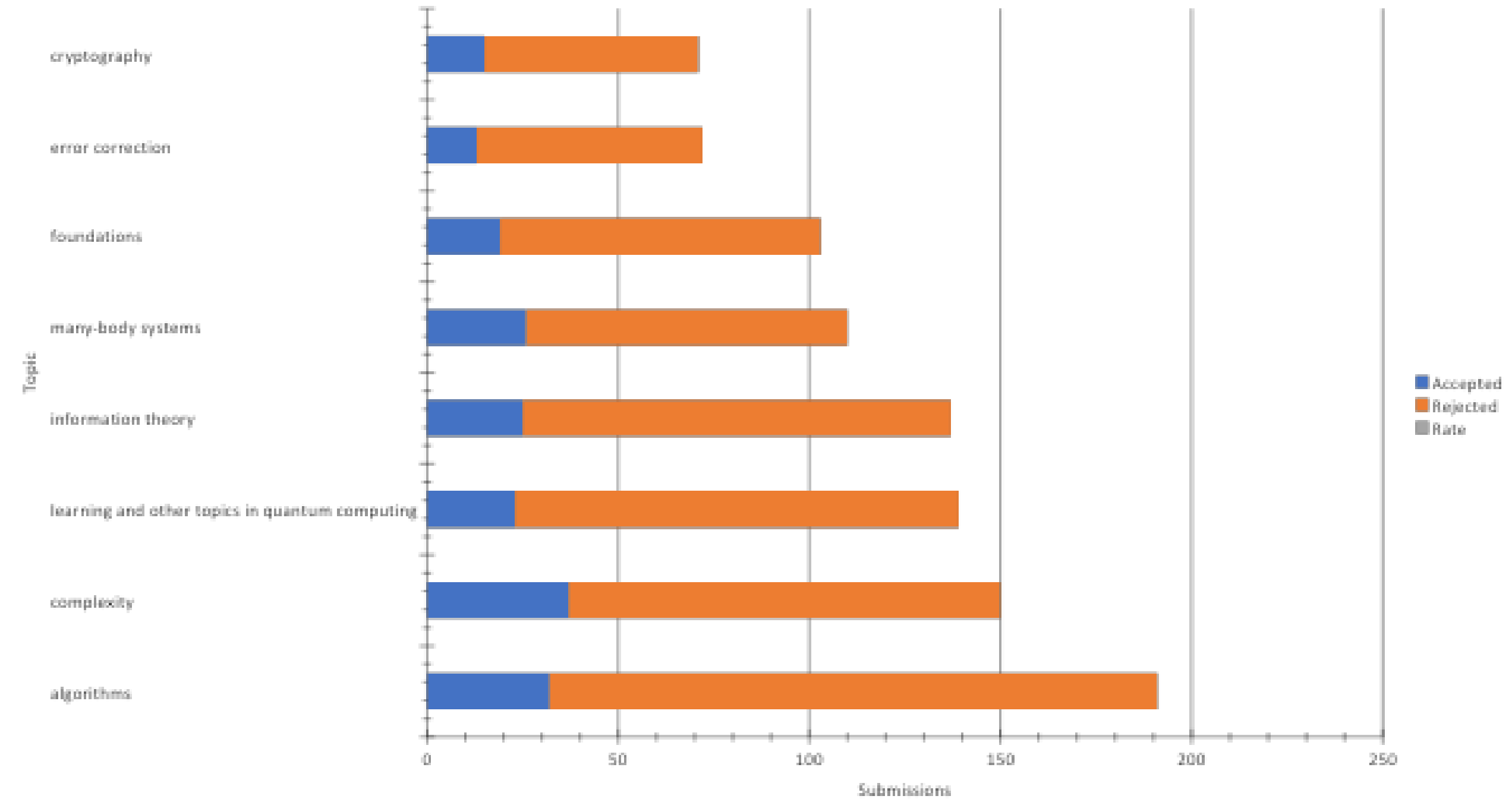
- 575 submissions to talk track (QIP' 23: 558, QIP'22: 514, QIP'21: 462)
- 111 accepted submissions (109 talks due to 2 merges)
- 96 regular talks
- 10 short plenaries
- 3 long plenaries (also, 3– 2 invited plenaries from SC)
- acceptance rate: 19.3% (QIP '23: 19.9%, QIP'22: 22.0%, QIP'21: 24.0%)

249 poster only submissions. (Almost) All rejected talks accepted as posters. 402 posters signed up to present. 3 “reject even as poster”

Number of Submissions by Topic

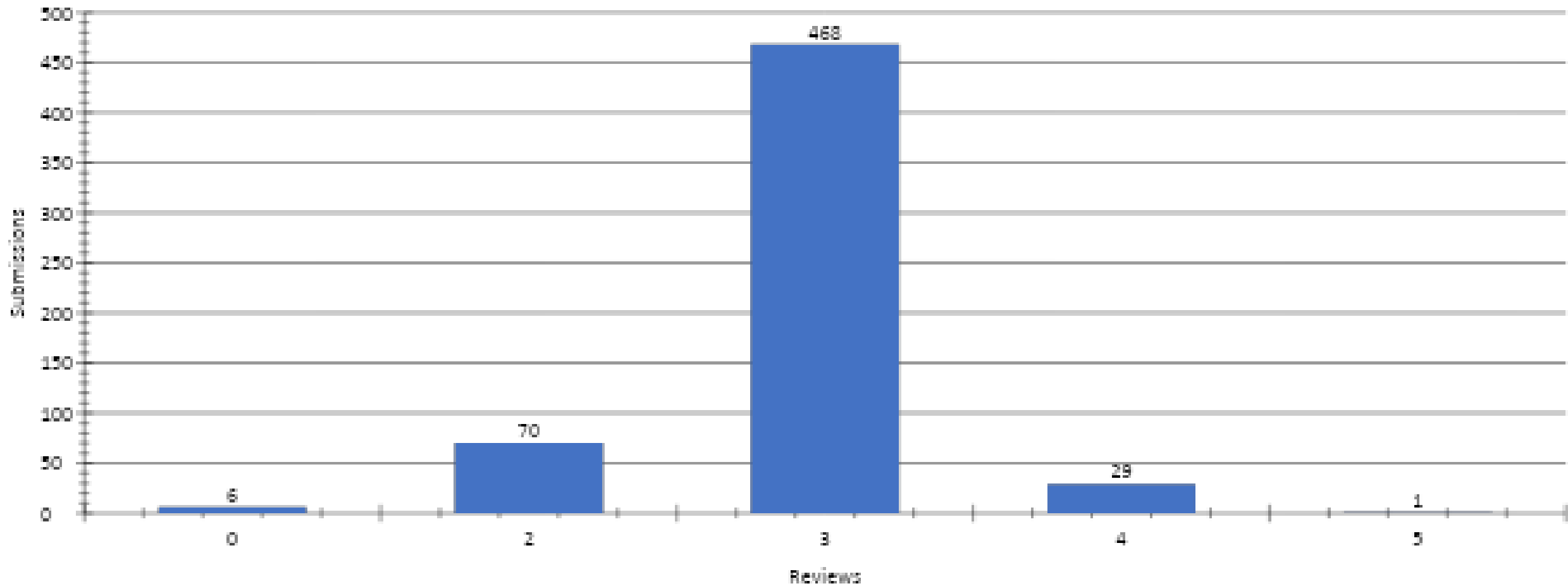


Number of Submissions and Acceptance Rate by Topic



1666 reviews, 514 external reviewers, 769 external reviews

Number of Reviews per Submission



Process

- PC members bid on and receive assignment of ~28 submissions a few days after deadline
- 4 weeks later: internal deadline for completing reviews
- Initially propose accept ~10, propose reject ~200 where all reviews are definite and agree.
- Week 1: Flag papers with diverging reviews, reach agreement where possible, add more voices, seek further reviews. Discuss and finalize first proposals of accept/reject.
- Week 2: make comparisons within subfields, determine relative merits of similar papers. Discuss and finalize further 200 to reject, 30 to accept.
- Week 3: 40 accepted, 400 rejected. 130 papers remain for 70 slots. Compare across subfields. Vote on plenaries.
- Week 4: Championing and voting on remaining papers. Voting on best student paper.
- Week 5: decisions finalized by Nov 14, reviews cleaned up. Decisions sent out by Nov 16.

Best Student Paper Award

Congratulations Allen Liu, MIT!

428 Learning quantum Hamiltonians at any temperature in polynomial time
(with Ainesh Bakshi, Ankur Moitra and Ewin Tang)

Thursday Plenary, 9am

The future

- If number of submissions continues to grow, this model won't scale
- Probably would have to break into tracks with minimal/no interactions between tracks handled by "area chairs". PC members probably belong to fixed track.
- Perhaps then area chairs negotiate to blend their respective lists
- Investigate using system other than easychair
- Down sides: papers bridging tracks may be disadvantaged, group-think enhanced when only info theorists look at info theory, only complexity theorists look at complexity theory.

From call for papers: "We aim to ensure the broadest selection of talks at all conferences in our field. In light of the large volume of high-quality research being produced and the low acceptance rates at recent editions of QIP, submissions that have already been presented at another conference of a similar scope to QIP (eg TQC) are discouraged."

Program committee of 66 THANK YOU!

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We hope you enjoy the program!

2024 TAIPEI



QIP 2025

Margo Ginsberg
(Duke University)

2024 TAIPEI

28th Annual Conference on Quantum Information Processing

22-28 February 2025
Raleigh, North Carolina, USA



Raleigh, North Carolina + Research Triangle Park

3 Research Universities: NC State University, Duke University, UNC Chapel Hill (also ARO office)

Industry: IBM, Cisco, SAS, NetApp/EMC2, Nvidia, Qualcomm, RedHat, ...

Durham/Raleigh a humid subtropical climate with mild winters and total monthly precipitation of 3.5".

February average daily high is 56°F (13 C), average daily low is 34°F (1 C).



Raleigh Convention Center (RCC)

Meeting Facility: 46,450sqm (500,000 sf)

Largest Ballroom: 2970sqm (32,000 sf)

Classroom Capacity: 2440

of Rooms: 16

Adjacent to 3 hotels

25.4km (15.8 mi) from RDU Int'l Airport



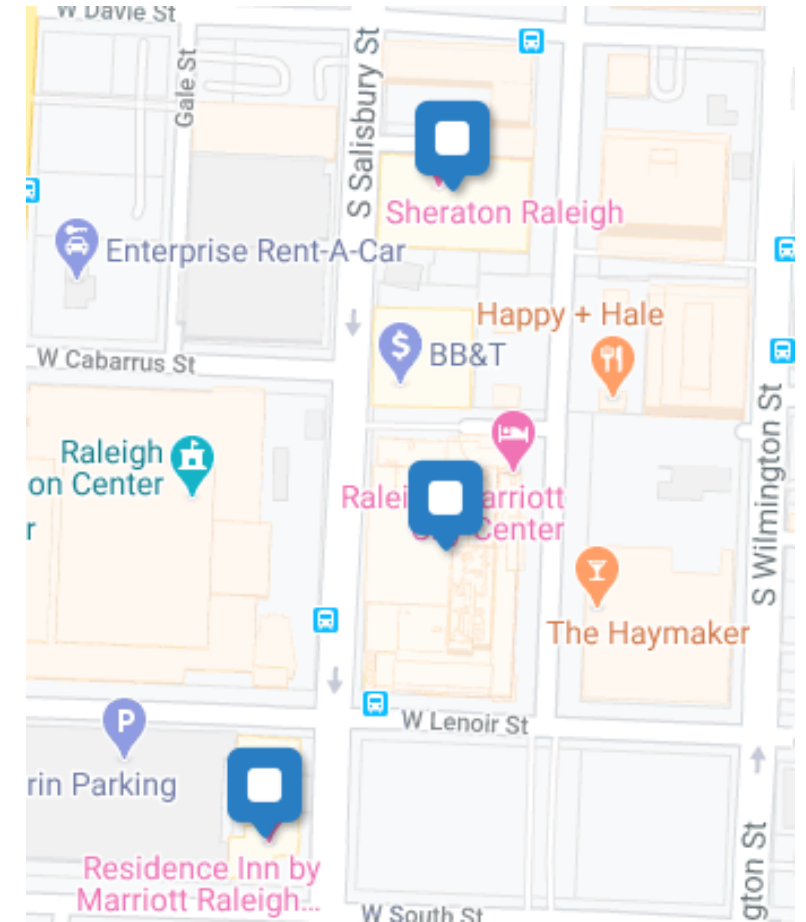
Downtown Raleigh Hotels

Negotiated group rates \$169-222/night

928 sleeping rooms at three hotels within <0.25 miles of Convention Center.

4th property to open in 2024

Over 80 restaurants and food establishments in downtown business district



International Air Access

Raleigh-Durham Int'l (RDU)

- Serviced by all major US carriers
- Int'l Carriers: American, Air Canada, Delta
- Int'l Non-Stop: Cancun, Frankfurt, London (Heathrow), Mexico City, Montreal, Paris (de Gaulle), Reykjavik, Toronto
- Int'l 1 stop flights: Sydney via LAX; Shanghai via O'Hare, Newark and San Francisco; Tokyo via JFK, Boston and Seattle; Singapore via San Francisco, Newark and Seattle

Charlotte Douglas Int'l (CLT)

- 2.5 hr drive
- American Airlines hub
- Int'l Non-Stop: Dublin, Madrid, Munich and more.



Ground Transportation

Approximately 25 minutes

Taxi \$50

Lyft & Uber

Limo \$50 or Rental Cars



THE UNIVERSITY
of NORTH CAROLINA
at CHAPEL HILL

Airport

Raleigh

NC STATE UNIVERSITY

IBM Quantum Innovation Center



From the Mountains to the Sea: North Carolina Impressions

Local Organizing Committee

Duke University

Iman Marvian

Kenneth Brown

Margo Ginsberg, Staff

North Carolina State University

Frank Mueller

Daniel Stancil

Mary Catherine Cole, Staff

See you next year in North America

Get in touch

qip2025@duke.edu

Coming soon: www.qip2025.duke.edu





Bids for future QIPs: Singapore and Latvia

QIP 2026 in Singapore

- Tropical climate: temperature 28° and humidity 83% all year
- Prime location and travel hub with worldwide direct flights and open to all with an efficient and inclusive visa process.
- Efficient infrastructure and a variety of venues for hosting large conferences (flexible in terms of number of parallel sessions, participant number)
- CQT established in 2007 with over 250+ CQTians from 30+ countries
- Strong local support with a team experienced in organising large conferences (QIP'11, QCrypt'12, TQC'14, QCMC'16, AQIS'17).
- Vibrant local research hub and quantum ecosystem.



QIP 2026 in Riga, Latvia



Venue: Atta Centre.

Admin support: Latvian Quantum Initiative





Discussion: Future of QIP

2024 TAIPEI

QIP is growing significantly

- Packed schedule (9am-9pm). More tracks?
- Declining acceptance rate
- Increased workload (esp. on PC chairs)

Solutions?

- Another conference of a similar caliber? TQC?
Summer/winter QIP.
- Different selection (PC) procedure/structure.
PC chair -> Area chairs

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